Conditional entropy is the entropy that remains in one variable once the other is known. It never exceeds the entropy of the variable on its own, and it equals that entropy only when the two variables are independent. A weather report that always agrees with the weather leaves no conditional entropy at all, while a report written at random leaves all of it.

The chain rule ties these quantities together. The joint entropy of two variables is the entropy of the first plus the conditional entropy of the second given the first. Applied again and again, the rule splits the entropy of a long message into the cost of each symbol given the symbols before it.

Students are told early in the course that a thing which is self-evident to a theorist may still need a proof, and the proof of the chain rule is short enough to give in full. It follows from the definition of conditional probability and the additivity of the logarithm, and it needs nothing else.

Mutual information is the difference between an entropy and a conditional entropy. It is symmetric, it is never negative, and it is zero exactly when the two variables are independent. Every later result in this book about what a channel can carry is a statement about this one quantity.